

RAHMAN HUSSAIN

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WORK EXPERIENCE

SUNY Research Foundation

Research Aide

Apr 2026 - May 2026

- Designed a two-tier architecture for a log anomaly detector that uses a fast small language model to screen most traffic and escalates uncertain cases to a second-stage model for deeper review.
- Fine-tuned a small language model to flag anomalies in two production log systems, hitting F1 scores above **97%** and **90%**.

Youro

Software Engineer Intern

Aug 2025 - Dec 2025

- Owned full-stack development of a patient-facing urology telehealth platform, from authentication and appointment booking to doctor-patient chat and medication tracking, and shipped it to production in **4 months**.
- Partnered with the founding urologist to design an adaptive intake questionnaire that compiles patient answers into a pre-consult report, cutting review time from **~15 minutes to under 4**.
- Rebuilt the API client layer with request deduplication, response caching, and refresh-token rotation, cutting redundant backend calls **30%** and closing a token-replay security gap.
- Built an AI diagnosis-assist tool that uses an LLM to pull symptoms from intake reports and a RAG layer trained on clinical guidelines to rank the top 3 likely diagnoses, which included the urologist's final diagnosis on **85%** of consults.

Goodz Pvt. Ltd

Software Engineer

Jul 2023 – Dec 2024

- Engineered the core order service that handles placement, confirmation, dispatch, and delivery for **4,000+ orders a day** on Node.js and PostgreSQL, and hardened it with idempotent transitions to cut stuck-order rates **40%**.
- Owned product development for vendor payouts and reconciliation, replacing a manual spreadsheet workflow with an automated nightly job that cut reconciliation from **2 days to 20 minutes**.
- Traced a production outage during a 20x holiday traffic spike to a saturated database connection pool and fixed it with a Redis caching layer, cutting p99 latency **97%** with zero downtime.
- Designed and built a custom high-performance data grid for 50,000+ order records, cutting rendering lag **73%**.

PROJECTS

L-GRIP | Independent research | Python, PyTorch, Llama, Kafka, Neo4j

Sept 2026

- Redesigned the anomaly filter into a multi-model pipeline and benchmarked it against a single 70B model on the same hardware, cutting inference time **78%** and memory use **82%** per log line.
- Built a **Kafka** pipeline that splits incoming logs into two parallel paths, one writing to a graph database and one running anomaly detection, and added a sync check so detection never reads the graph before the matching write lands.

Image Indexing & Semantic Search | UB Research | Python, FastAPI, FAISS, CLIP

Mar 2026

- Indexed a client's photo archive using InsightFace for face tags and RAM for object tags, keeping CPU load low.
- Built a hybrid search pipeline that pre-filters candidates in SQLite before running CLIP and FAISS vector search.

HealthAI Connect | Python, Flask, LangChain, FAISS

Aug 2025

- Built a multi-modal diagnostic platform with a RAG chatbot over ~9,000 records, hitting **98.5% accuracy**.
- Published in Taylor & Francis *AI Frontiers*.

SKILLS

- Backend Tech:** Python, Node.js, FastAPI, Flask, Spring Boot, REST APIs, WebSockets, SQLAlchemy
- Frontend Tech:** React, TypeScript, Tailwind CSS
- Infrastructure & Databases:** PostgreSQL, Docker, AWS (S3, CloudFront), Kafka, Redis, MySQL, Neo4j, Git
- Testing:** pytest, JUnit, Jest, RTL
- AI & ML:** PyTorch, LangChain, RAG, FAISS (vector search), recommender systems

EDUCATION

University at Buffalo, SUNY | Master of Science in Computer Science and Engineering

May 2026

Osmania University | Bachelor of Engineering in Information Technology

Jun 2024